MTH2001

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Analysis

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Other examiners: Dr Valentina Di Proietto, Dr Layal Hakim

Duration: 2 HOURS.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Consider the following subset of the real numbers

$$S = (-\infty, 5) \cup \{2020\}.$$

(i) Is the set S open? Is the set S closed? In each case justify your answer.

(ii) Determine the set of limit points of S and the set of interior points of S. (10)

(b) Prove, using the ϵ definition of limit, that

$$\lim_{x\to\infty} \frac{1}{x^3 + 2x + 1} = 0.$$

(10)

(c) Let $A \subset \mathbb{R}^n$, and $f, g : A \to \mathbb{R}$, and let

$$\lim_{x\to a} f(x) = f(a), \quad \lim_{x\to a} g(x) = g(a).$$

Prove, using the standard definition of convergence in $\mathbb{R}^n$, that

$$\lim_{x\to a}(f(x) + g(x)) = f(a) + g(a).$$

(10)

(d) (i) State the Inverse Function Theorem for a function

$$g : \mathbb{R}^n \to \mathbb{R}^n.$$

(8)

(ii) Let $h : \mathbb{R}^2 \to \mathbb{R}^2$ be given by

$$h(x, y) = \begin{pmatrix} x + 5y^5 \\ 7xy - y \end{pmatrix}.$$

Compute the partial derivatives $D_i h^j(p)$ for $i, j = 1, 2$. Then, using the Inverse Function Theorem, show that for all p in a neighbourhood of $(0, 0)$ the equation $h(p) = q$ has a solution.

(12)

[**50**]

SECTION B

2. We consider the function

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

where $a_i \in \mathbb{R}$ for all $i = 0, \ldots, n-1$.

(a) Prove that $p(x)$ is bounded on $\mathbb{R}$ if and only if $n = 0$. (7)

(b) If n is odd prove that for all $\alpha \in \mathbb{R}$ there exists $x_0 \in \mathbb{R}$ such that $p(x_0) = \alpha$. (8)

(c) If n is even prove that there exist x_0 and x_1 such that $x_0 \neq x_1$ and such that $p(x_0) = p(x_1)$. (10)

[**25**]

3. (a) State the Riemann criterion for integrability. (5)

(b) Let $f : [a, b] \to \mathbb{R}$ be an increasing function, and let partition P be given by

$$P = \left(a + \frac{i(b-a)}{n}\right)_{i=0,\ldots,n} = \left(a, a + \frac{b-a}{n}, a + \frac{2(b-a)}{n}, ..., b\right).$$

Define the upper and lower Riemann sums of f over partition P, and use them to show that f is Riemann-integrable on $[a, b]$. (8)

(c) Compute the limit

$$\lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} \sin\left(\frac{k\pi}{2n}\right)}{n}.$$

(12)

[**25**]

4. Let λ be a real number such that $\lambda \in \left(-\frac{e}{2}, \frac{e}{2}\right)$. We consider the function

$$f(x) = e^x - \lambda x^2.$$

(a) Let $g(x) = f'(x)$. Prove that $g(x)$ is Lipschitz continuous on $[0, 1]$ and compute the smallest Lipschitz constant L for $g(x)$ on $[0, 1]$. (8)

(b) Compute

$$\lim_{x \to \infty} f(x)$$

and

$$\lim_{x \to -\infty} f(x).$$

(8)

(c) How many solutions has the equation

$$f(x) = 0?$$

Justify your answer. (9)

[**25**]